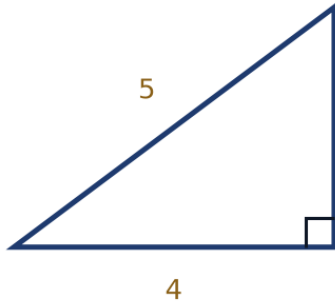


Advanced Trigonometry



Trigonometric identities: Pythagorean and reciprocal

- 1 Given $\cos \theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin \theta$ and $\tan \theta$.



Sum and difference identities

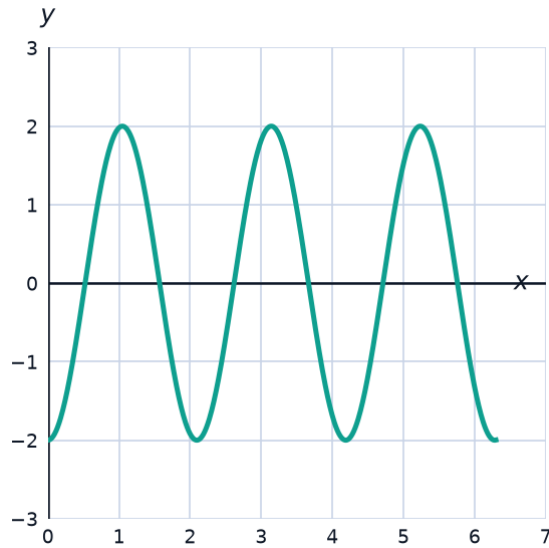
- 2 Use the sum identity to find the exact value of $\sin(75^\circ)$, using $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$.

Double angle identities

- 3 Given $\sin \theta = \frac{3}{5}$ and θ in the first quadrant, find $\sin(2\theta)$ and $\cos(2\theta)$.

Graphs and transformations of trigonometric functions

- 4 State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.



Inverse trigonometric functions

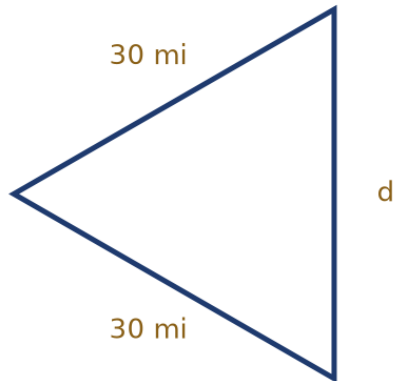
- 5 Find the exact value of $\arcsin\left(\frac{1}{2}\right)$.
- 6 Find the exact value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$.

Trigonometric equations with multiple solutions

- 7 Solve $2\cos^2\theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

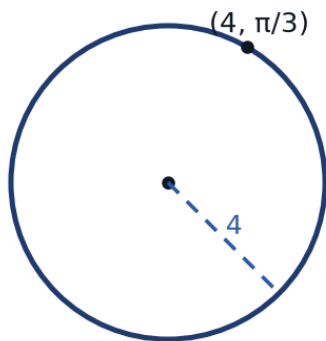
Applications: Law of Sines and Law of Cosines together

- 8 Two ships leave a port at the same time, one traveling at 20° and the other at 80° (measured from north), each at 15 mph. Find the distance between the ships after 2 hours, to the nearest mile, using the Law of Cosines.



Polar coordinates

- 9 Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.



A multi-part trigonometric identity verification problem

- 10 This question has two parts. (a) Use the identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan \theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan \theta = 3$.

Half-angle identities

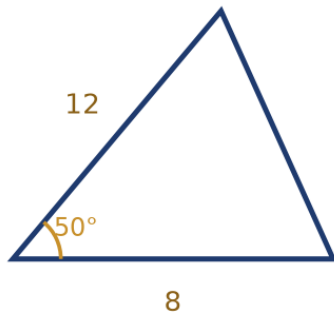
- 11 Use the half-angle identity to find $\cos(15^\circ)$, given $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.
-

Solving trigonometric equations that factor

- 12 Solve $2\sin^2 \theta - \sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
-

Area of a triangle using trigonometry

- 13 Find the area of a triangle with sides 8 and 12, and included angle 50° , using $\text{Area} = \frac{1}{2}ab\sin C$.

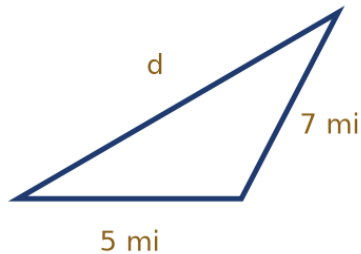


Simplifying trigonometric expressions using identities

- 14 Simplify $\sin^2 \theta + \cos^2 \theta + \tan^2 \theta$ in terms of $\sec \theta$.
-

A multi-part navigation and triangulation problem

- 15 This question has three parts. A hiker walks 5 miles due east, then turns and walks 7 miles at a bearing of 40° north of east. (a) Find the angle between the two legs of the hike (measured as the interior angle at the turning point). (b) Use the Law of Cosines to find the straight-line distance from the start to the end point, to the nearest tenth of a mile. (c) Explain why this distance is less than the total distance walked (12 miles).



Verifying a trigonometric identity

- 16 Verify the identity $\frac{1 - \cos^2 \theta}{\sin \theta} = \sin \theta$.

Multiple choice — easy

- 17 Given $\sin \theta = \frac{15}{17}$ and θ in the second quadrant, find $\cos \theta$.

- a. $-\frac{15}{17}$
- b. $\frac{15}{17}$
- c. $-\frac{8}{17}$
- d. $\frac{8}{17}$

- 18 Given $\sin \theta = -\frac{11}{61}$ and θ in the fourth quadrant, find $\cos \theta$.

- a. $\frac{11}{61}$
- b. $\frac{60}{61}$
- c. $-\frac{60}{61}$
- d. $-\frac{11}{61}$

19 Given $\cos \theta = -\frac{4}{5}$ and θ in the second quadrant, find $\sin \theta$.

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $-\frac{3}{5}$

20 Given $\cos \theta = -\frac{12}{37}$ and θ in the second quadrant, find $\sin \theta$.

a. $\frac{35}{37}$

b. $-\frac{35}{37}$

c. $-\frac{12}{37}$

d. $\frac{12}{37}$

21 Given $\cos \theta = -\frac{7}{25}$ and θ in the second quadrant, find $\sin \theta$.

a. $\frac{24}{25}$

b. $-\frac{24}{25}$

c. $\frac{7}{25}$

d. $-\frac{7}{25}$

22 Given $\cos \theta = \frac{12}{37}$ and θ in the fourth quadrant, find $\sin \theta$.

a. $-\frac{12}{37}$

b. $\frac{12}{37}$

c. $\frac{35}{37}$

d. $-\frac{35}{37}$

23 Given $\cos \theta = -\frac{12}{13}$ and θ in the third quadrant, find $\sin \theta$.

a. $-\frac{12}{13}$

b. $\frac{12}{13}$

c. $-\frac{5}{13}$

d. $\frac{5}{13}$

24 Given $\cos \theta = \frac{12}{37}$ and θ in the first quadrant, find $\sin \theta$.

a. $\frac{12}{37}$

b. $\frac{35}{37}$

c. $-\frac{35}{37}$

d. $-\frac{12}{37}$

25 Given $\sin \theta = -\frac{35}{37}$ and θ in the fourth quadrant, find $\cos \theta$.

a. $-\frac{12}{37}$

b. $\frac{35}{37}$

c. $\frac{12}{37}$

d. $-\frac{35}{37}$

26 Given $\cos \theta = -\frac{8}{17}$ and θ in the second quadrant, find $\sin \theta$.

a. $\frac{8}{17}$

b. $-\frac{8}{17}$

c. $-\frac{15}{17}$

d. $\frac{15}{17}$

27 Given $\cos \theta = -\frac{24}{25}$ and θ in the second quadrant, find $\sin \theta$.

- a. $\frac{7}{25}$
- b. $-\frac{24}{25}$
- c. $-\frac{7}{25}$
- d. $\frac{24}{25}$

28 Given $\cos \theta = -\frac{11}{61}$ and θ in the third quadrant, find $\sin \theta$.

- a. $\frac{11}{61}$
- b. $-\frac{60}{61}$
- c. $\frac{60}{61}$
- d. $-\frac{11}{61}$

29 Given $\cos \theta = -\frac{35}{37}$ and θ in the third quadrant, find $\sin \theta$.

- a. $\frac{12}{37}$
- b. $-\frac{35}{37}$
- c. $-\frac{12}{37}$
- d. $\frac{35}{37}$

30 Given $\cos \theta = \frac{24}{25}$ and θ in the first quadrant, find $\sin \theta$.

- a. $\frac{7}{25}$
- b. $-\frac{24}{25}$
- c. $-\frac{7}{25}$
- d. $\frac{24}{25}$

31 Given $\cos \theta = \frac{12}{13}$ and θ in the first quadrant, find $\sin \theta$.

a. $-\frac{5}{13}$

b. $-\frac{12}{13}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

Multiple choice — medium

32 Solve $2\sin \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

a. 0°

b. $210^\circ, 330^\circ$

c. 180°

d. $120^\circ, 240^\circ$

33 Solve $2\cos \theta = -\sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $45^\circ, 315^\circ$

b. $150^\circ, 210^\circ$

c. $60^\circ, 120^\circ$

d. 0°

34 Solve $2\cos \theta = \sqrt{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $225^\circ, 315^\circ$

c. $45^\circ, 315^\circ$

d. $45^\circ, 135^\circ$

35 Solve $\sin \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

a. 270°

b. $150^\circ, 210^\circ$

c. $60^\circ, 300^\circ$

d. 0°

36 Solve $\sin \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $225^\circ, 315^\circ$

b. $45^\circ, 135^\circ$

c. 90°

d. 0°

37 Solve $2\cos \theta = \sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $30^\circ, 330^\circ$

b. $60^\circ, 300^\circ$

c. 0°

d. $120^\circ, 240^\circ$

38 Solve $2\sin \theta = -\sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $30^\circ, 330^\circ$

b. $240^\circ, 300^\circ$

c. $120^\circ, 240^\circ$

d. $60^\circ, 300^\circ$

39 Solve $2\sin \theta = \sqrt{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $45^\circ, 135^\circ$
- b. $150^\circ, 210^\circ$
- c. 0°
- d. $135^\circ, 225^\circ$

40 Solve $2\cos \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

- a. $240^\circ, 300^\circ$
- b. $45^\circ, 135^\circ$
- c. $150^\circ, 210^\circ$
- d. $60^\circ, 300^\circ$

41 Solve $2\sin \theta = -\sqrt{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $120^\circ, 240^\circ$
- b. $45^\circ, 135^\circ$
- c. 0°
- d. $225^\circ, 315^\circ$

42 Solve $\cos \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

- a. $45^\circ, 135^\circ$
- b. 180°
- c. $150^\circ, 210^\circ$
- d. $30^\circ, 150^\circ$

43 Solve $2\sin \theta = \sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $60^\circ, 120^\circ$

b. 180°

c. 90°

d. $30^\circ, 330^\circ$

44 Solve $2\cos \theta = -\sqrt{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $45^\circ, 315^\circ$

b. $135^\circ, 225^\circ$

c. 0°

d. 180°

45 Solve $\cos \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $210^\circ, 330^\circ$

b. 0°

c. 270°

d. $30^\circ, 330^\circ$

46 Solve $2\cos \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. 180°

c. $120^\circ, 240^\circ$

d. $150^\circ, 210^\circ$

Multiple choice — hard

47 Given $\sin \theta = \frac{15}{17}$, find $\cos(2\theta)$.

a. $\frac{240}{289}$

b. $-\frac{161}{289}$

c. $\frac{161}{289}$

d. $-\frac{240}{289}$

48 Given $\sin \theta = \frac{20}{29}$, find $\cos(2\theta)$.

a. $\frac{840}{841}$

b. $-\frac{41}{841}$

c. $\frac{41}{841}$

d. $-\frac{840}{841}$

49 Given $\sin \theta = \frac{24}{25}$, find $\cos(2\theta)$.

a. $-\frac{527}{625}$

b. $\frac{527}{625}$

c. $\frac{336}{625}$

d. $-\frac{336}{625}$

50 Given $\sin \theta = \frac{8}{17}$, find $\cos(2\theta)$.

a. $-\frac{240}{289}$

b. $\frac{161}{289}$

c. $-\frac{161}{289}$

d. $\frac{240}{289}$

51 Given $\sin \theta = \frac{5}{13}$, find $\cos(2\theta)$.

a. $-\frac{120}{169}$

b. $\frac{119}{169}$

c. $\frac{120}{169}$

d. $-\frac{119}{169}$

52 Two sides of a triangle measure 7 and 8, with an included angle of 120° . Find the length of the third side, using the Law of Cosines.

a. 14

b. 13

c. 12

d. 11

53 Two sides of a triangle measure 16 and 16, with an included angle of 60° . Find the length of the third side, using the Law of Cosines.

a. 23

b. 16

c. 15

d. 17

54 Two sides of a triangle measure 11 and 11, with an included angle of 60° . Find the length of the third side, using the Law of Cosines.

a. 16

b. 10

c. 12

d. 11

- 55 Two sides of a triangle measure 3 and 3, with an included angle of 60° . Find the length of the third side, using the Law of Cosines.
- a. 3
 - b. 4
 - c. 2
 - d. 5
- 56 Two sides of a triangle measure 3 and 8, with an included angle of 60° . Find the length of the third side, using the Law of Cosines.
- a. 8
 - b. 7
 - c. 6
 - d. 9
- 57 Find the area of a triangle with sides 11 and 5, and included angle 150° , using $A = \frac{1}{2}ab\sin C$.
- a. $\frac{55}{2}$
 - b. $\frac{59}{4}$
 - c. $-\frac{55}{4}\sqrt{3}$
 - d. $\frac{55}{4}$
- 58 Find the area of a triangle with sides 12 and 12, and included angle 150° , using $A = \frac{1}{2}ab\sin C$.
- a. 72
 - b. $-36\sqrt{3}$
 - c. 37
 - d. 36

- 59 Find the area of a triangle with sides 13 and 7, and included angle 90° , using $A = \frac{1}{2}ab\sin C$.
- a. $\frac{93}{2}$
 - b. $\frac{91}{2}$
 - c. 0
 - d. 91
- 60 Find the area of a triangle with sides 14 and 9, and included angle 150° , using $A = \frac{1}{2}ab\sin C$.
- a. $-\frac{63}{2}\sqrt{3}$
 - b. 63
 - c. $\frac{63}{2}$
 - d. $\frac{65}{2}$
- 61 Find the area of a triangle with sides 12 and 4, and included angle 150° , using $A = \frac{1}{2}ab\sin C$.
- a. 24
 - b. 13
 - c. 12
 - d. $-12\sqrt{3}$